

Jason Jaeseung Kim

Biomedical Engineering Undergraduate | Deep Learning for Medical Applications

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RESEARCH FOCUS

Through medical-imaging deep-learning research, I became interested not only in preprocessing and quantitative evaluation, but also in how model outputs can be interpreted and used with real clinical imaging data. Building on my experience with 3D MRI, I aim to extend this work to diverse clinical imaging problems.

EDUCATION

Yonsei University - B.S. Candidate in Biomedical Engineering

Mar 2021 - Expected Feb 2027

Mirae Campus | Second Major in Electrical and Electronic Engineering, Sinchon Campus

GPA	4.50 Scale	Percentile	Department Rank
4.03 / 4.30	4.18 / 4.50	97.90 / 100	1 / 49

Selected coursework: Image Processing and Applications; Digital Signal Processing; Signals and Systems; Deep Learning Laboratory; Public Health and Medical Big Data Analysis.

RESEARCH EXPERIENCE

Former Undergraduate Researcher, Advanced Image Processing Techniques Lab, Yonsei University

Dec 2023 - Feb 2026

Advisor: Prof. Joon Yul Choi | Mirae Campus

- Preparing two first-author journal manuscripts in quantitative MRI and medical image deep learning.
- Designed and evaluated a 3D MRI image-to-image regression approach for quantitative T1 estimation, comparing input conditions, preprocessing and harmonization strategies, and model configurations.
- Built automated 3D medical-image pipelines for registration, masking, resampling, training, baseline reproduction, and tissue-wise evaluation.
- Conducted an SPM12 voxel-wise quantitative MRI analysis of overwork, including cohort curation, covariate adjustment, explicit masking, FWE correction, atlas-based reporting, and automated QC.
- Contributed to drafting a multi-year government-funded R&D; proposal and literature review; prepared conference abstracts, oral presentations, figures, and tables.
- Ongoing graduation research: extending NAMER by evaluating multiple fastMRI subjects and revising CNN training so the image prior better supports downstream motion estimation.

Undergraduate Researcher, Bio-Metamaterials Lab, Yonsei University

Jul 2022 - Aug 2022

Advisor: Prof. Seung Ho Choi | Mirae Campus

- Participated in the initial installation and setup of an ellipsometer and analyzed the compositional and optical properties of biomaterials.
- Built a prototype tensile-strength measurement device and implemented a MATLAB-based user interface for measurement verification.

SELECTED RESEARCH OUTPUTS

- Oral Presentation and Outstanding Paper Award for the quantitative T1 estimation study, Korean Society of Medical and Biological Engineering Fall Conference, Nov 2025.
- Poster Presentation, NEURO2026, Jul 2026.

PROJECTS

Teacher-Difficulty-Calibrated Early-Exit Network

Undergraduate Course Project, 2026

- Undergraduate Deep Learning Laboratory course project: developed an image-classification network using knowledge distillation, teacher-difficulty guidance, dynamic early exits, and branch-fused deployment for efficient inference.

Project Lead, CareProofRecord - Deep-Learning-Based Medical Explanation Video Platform

Jun 2025 - Apr 2026

- Led the planning and system architecture of patient-facing medical explanation videos designed to improve understanding and reduce information-transfer errors.
- Designed a long-context generation workflow to mitigate hallucination risk, secured KRW 1,000,000 in university startup support, and explored clinical use with regional medical institutions.

TECHNICAL SKILLS

Programming	Python; MATLAB
Medical Imaging Software	SynthStrip (skull stripping); FSL FAST (tissue segmentation); ANTs (registration and resampling); SPM12 (voxel-wise GLM and FWE correction); FreeSurfer (ROI extraction)

MERIT-BASED SCHOLARSHIPS

- National Science and Engineering Scholarship, Korea Student Aid Foundation - full tuition (Spring 2023, Fall 2023, Spring 2026).
- Hankyungbum Scholarship, Cheongpa-Hankyungbum Scholarship Foundation, Yonsei University - full tuition (Spring 2022, Fall 2022).
- Academic Excellence Scholarship, Yonsei University (Fall 2021).

HONORS AND AWARDS

- **Academic Excellence Award**, Yonsei University - Spring 2023
- **Academic Excellence Award**, Yonsei University - Fall 2022
- **Academic Honors Award**, Yonsei University - Fall 2021
- **Academic Excellence Award**, Yonsei University - Spring 2021
- **Grand Prize**, ASC Programming Contest - 2021
- **Excellence Award**, Wise Coding Life Algorithm Competition, Yonsei University - 2022
- **Grand Prize**, MEDICI+ Startup Competition, Yonsei University Wonju LINC+ Project Group - 2021
- **Honorable Mention**, WAVE-LAB Startup Competition - 2025
- **Encouragement Award**, Post Corona-19 Startup Challenge - 2022